

# Structural Breaks

Chapter 6

Nicolas Cachanosky

Department of Economics and Finance

# Outline

- When the structure changes
- The Chow test
- The QLR and Bai-Perron test
- Unit roots or broken trends?
- How to react to structural breaks

# When the structure changes

Section 1

# When the rules change

- Remember that econometrics is a derivation of statistical application to natural sciences — constant and regular environment
  - Astronomy
  - Physics
  - Therefore, one can assume that the model's coefficients are constant across the sample (if the model is correctly specified)
- But...
  - Economic relationships (captured in the model's coefficients) emerge from the purposeful behavior of economic agents with subjective preferences and utility
  - When those relationships change, the “true” model coefficients do as well

# When the rules change

- **Stationarity** is the condition that a process has a stable probabilistic structure over time. Not that it is constant.
- Strict stationarity
  - A stochastic process  $\{y_t\}$  is **strictly stationary** if any joint distribution  $(y_{t_1}, \dots, y_{t_k})$  has the same joint distribution of  $(y_{t_1+h}, \dots, y_{t_k+h})$  for any  $h$
  - In simple terms, all observations come from the same probability distribution

# When the rules change

- Weak (covariance) stationarity
  - If data has similar probability properties
  - A stochastic process  $\{y_t\}$  is **weakly stationary** if:
    - (1)  $\mathbb{E}[y_t] = \mu < \infty \forall t$  — the mean exists and is finite
    - (2)  $Var(y_t) = \sigma^2 \forall t < \infty$  — the variance is finite and constant over time
    - (3)  $Cov(y_t, y_{t-h}) = \gamma(h) \forall t, h$  — The covariance between any two observations depends only on the lag  $h$ , not on when in time they occur.
  - Condition (1): Rules out trending means
  - Condition (2): Rules out changing variance
  - Condition (3): Memory is stable and time-independent

# When the rules change

- If a stationarity condition breaks, then there is a **structural break**
  - In statistical terms, you have two different series {processes} combined as one
- What happens if you pool two different series?
  - A single ARIMA over 1947-2019 GDP sample pools (a) the Korean War, (b) postwar productivity miracle, (c) the Great Moderation, and (d) the 2008 crisis followed by a slow recovery
  - Mean growth, persistence (AR terms), and shock variance (MA terms) are all different cross each period

# When the rules change

- Three ways a process breaks
  - **Mean shift.** The unconditional mean changes from  $\bar{y}_1$  to  $\bar{y}_2$  at date  $\tau$
  - **Variance break.** The volatility of innovations changes
  - **Coefficient break.** The dynamic structure changes
- These breaks can happen independently or together.
  - Great Moderation: Primarily a variance break
  - Volcker disinflation: A large mean shift in inflation and persistence

# When the rules change

- We are going to cover four structural break tests
  - Chow
  - QLR
  - Bai-Perron
  - Zivot-Andrews
- They all have something in common: Do a model that lets its coefficients change fit the data better than a single model for the whole sample?

# The Chow test

## Section 2

# The Chow test

- Assume you have a strong prior about structural break on  $t = \tau$
- Chow test (1960)
  - $y_t = x_t' \beta + \varepsilon_t$
  - H0:  $\beta$  is the same before and after  $\tau$
  - Ha:  $\beta$  is different before and after  $\tau$

# The Chow test

- Chow test application
  - Define  $D_t = 1\{t \geq \tau\}$
  - Restricted (R) model:  $y_t = x_t' \beta + \varepsilon_t$
  - Unrestricted (U) model:  $y_t = x_t' \beta_1 + (D_t x_t) \beta_2 + \varepsilon_t$ 
    - If
      - $D_t = 0 \rightarrow y_t = x_t \beta_1 + \varepsilon_t$
      - $D_t = 1 \rightarrow y_t = x_t (\beta_1 + \beta_2) + \varepsilon_t$
- Build your statistic
  - $F = \frac{(SSR_R + SSR_U)/k}{SSR_U/(T-2k)} \sim F(k, T-2k)$
  - If the difference between  $SSR_R$  and  $SSR_U$  is large enough, then  $\beta_1 \neq \beta_2$

# The Chow test

- Reading the Chow test
  - The test identifies a statistically significant difference between  $\beta_1$  and  $\beta_2$
  - It does not identify which coefficient(s) is (are) statistically different
  - Because  $\tau$  is given to the test, the the test cannot confirm that  $\tau$  is the correct break date (maybe is  $\tau - 2$ )
  - The test is testing for one structural break, the series may have more than one
  - It is not looking at changes in variance

# The Chow test

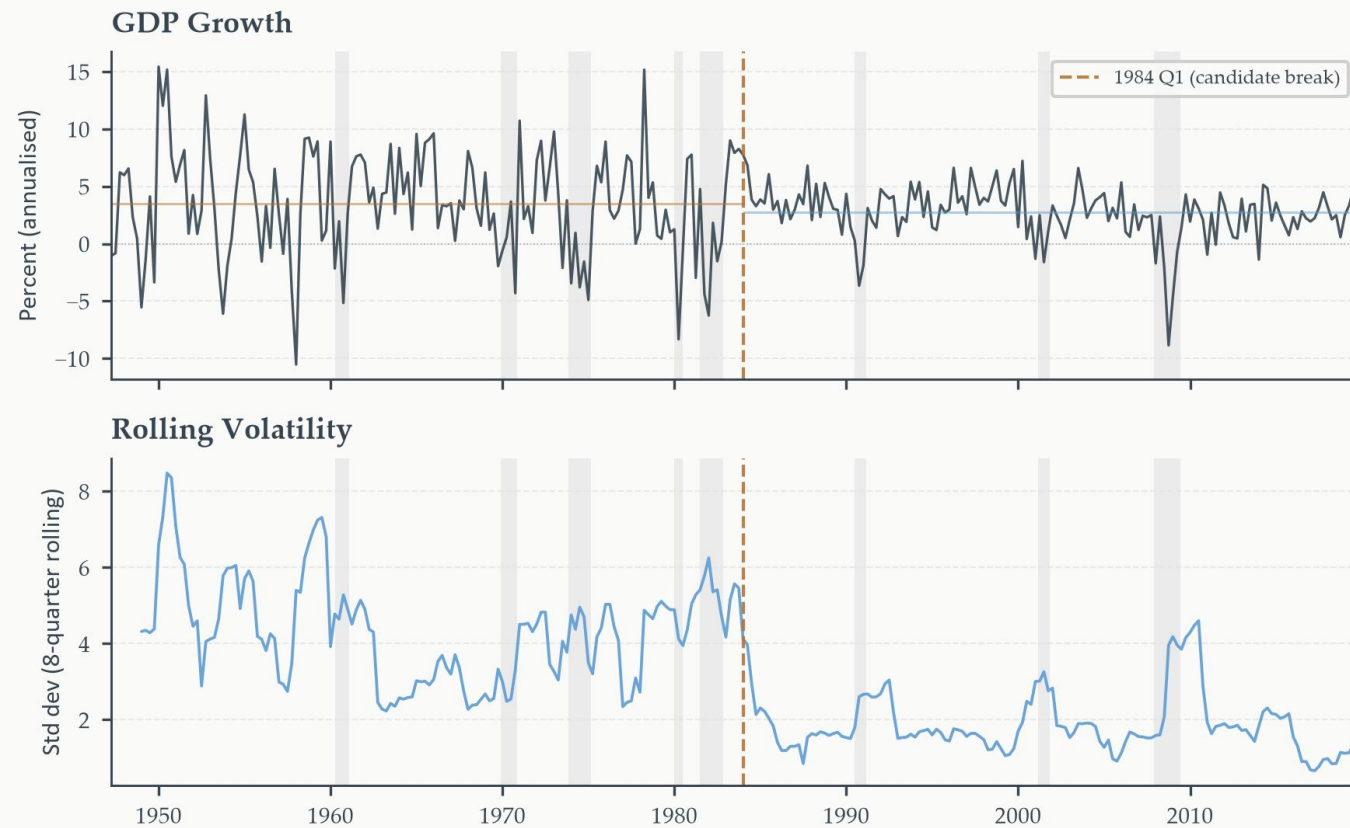
- The Great Moderation case

| Test                          | Statistic | p-value | Verdict       |
|-------------------------------|-----------|---------|---------------|
| Chow F-test (AR coefficients) | 1.023     | 0.383   | Do not reject |
| Variance F-test               | 4.733     | 0.000   | reject        |

The Great Moderation is a **variance break**, not a coefficient break

# The Chow test

US Real GDP Growth: The Great Moderation



# The Chow test

- The variance ratio test

- $F = \frac{S_{LARGE}^2}{S_{SMALL}^2} \sim F(n_{LARGE} - 1, n_{SMALL} - 1)$

# The QLR and Bai-Perron test

Section 3

# The QLR and Bai-Perron test

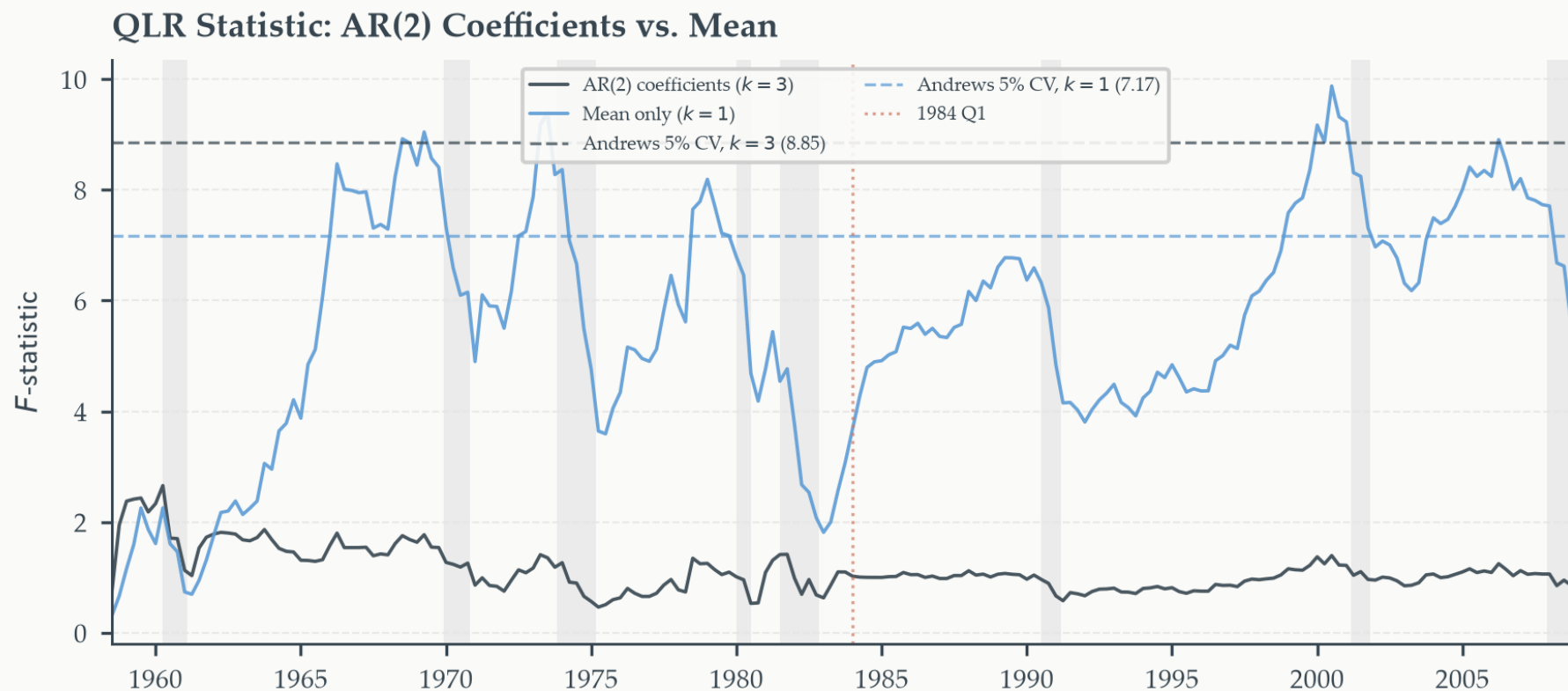
- Chow takes  $\tau$  as given
- QLR and Bai-Perron take  $\tau$  as unknown and use it as a parameter to be estimated
  - Idea: Compute the Chow  $F$ -statistic at **every** candidate date, and take the date that produces the maximum statistic (lowest p-value)

# The QLR and Bai-Perron test

- Quandt Likelihood Ration (QLR) test
  - $QLR = \sup_{\tau \in \tilde{\tau}} F(\tau)$
  - $\hat{\tau} = \operatorname{argmax}_{\tau \in \tilde{\tau}} F(\tau)$
  - Trimming:  $\tilde{\tau}$  excludes the first and last  $\pi_0$  (15%) of the sample
  - Estimate a Chow test for every  $\tau$ , the structural break happens at the  $\tau$  that produces the highest  $F$  statistic
- Davies Problem: Because  $\tau$  is unknown, the QLD test does not follow an  $F$  distribution
- Solution: Use Davies asymptotic values

# The QLR and Bai-Perron test

## Quandt Likelihood Ratio Test -- Break Date Search



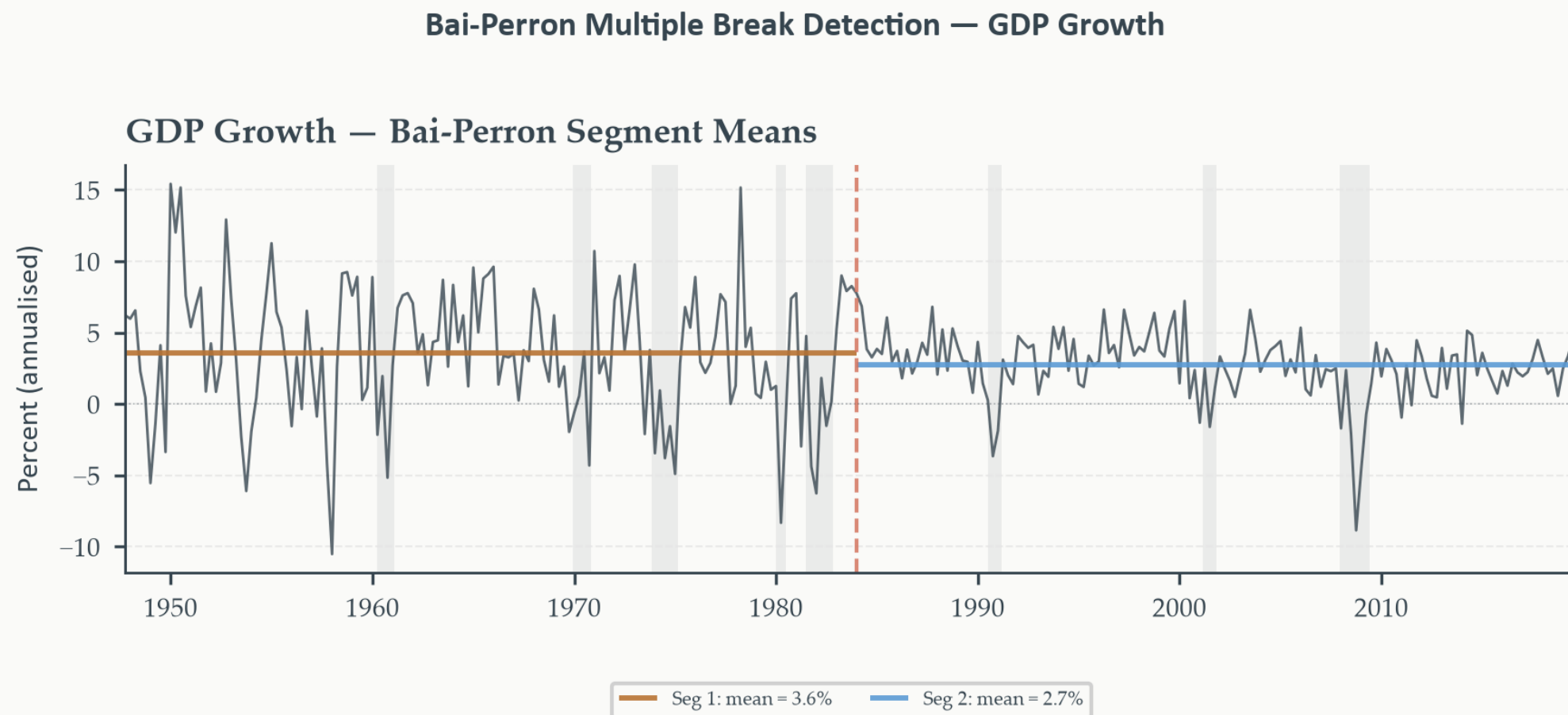
# The QLR and Bai-Perron test

- Bai-Perron test
  - QLR asks about one structural break
  - Bai-Perron ask about the presence of  $m \geq 1$  jointly unknown breaks
  - $(\hat{\tau}_1, \dots, \hat{\tau}_m) = \operatorname{argmin}_{\{\tau_1, \dots, \tau_m\}} \sum_{j=1}^{m+1} \sum_{t=\tau_{j-1}+1}^{\tau_j} (y_t - x_t' \beta_j)^2$

# The QLR and Bai-Perron test

- Bai-Perron test (cont...)
  - UDmax: Run a QLR test for each  $\tau = 1, \dots, m$  and combine them into one omnibus statistic asking “is there at least one break at all”?
  - WDmax: Same idea, but reweighted for equal power against each  $\tau$
  - Once there is evidence of a structural break, start with  $m = 1$  test if residuals (statistically) decrease. Increase  $m$  by 1 until residuals show no statistical difference

# The QLR and Bai-Perron test



# The QLR and Bai-Perron test

Bai-Perron Segment Statistics – AR(2) for GDP Growth

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| Segment | Start | End | Mean | Std Dev | Obs |
|---------|-------|-----|------|---------|-----|
|---------|-------|-----|------|---------|-----|

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|           |         |         |      |      |     |
|-----------|---------|---------|------|------|-----|
| Segment 1 | 1947 Q4 | 1984 Q1 | 3.58 | 4.68 | 146 |
|-----------|---------|---------|------|------|-----|

|           |         |         |      |      |     |
|-----------|---------|---------|------|------|-----|
| Segment 2 | 1984 Q1 | 2019 Q4 | 2.72 | 2.29 | 144 |
|-----------|---------|---------|------|------|-----|

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Mean and std dev in annualised percentage points.

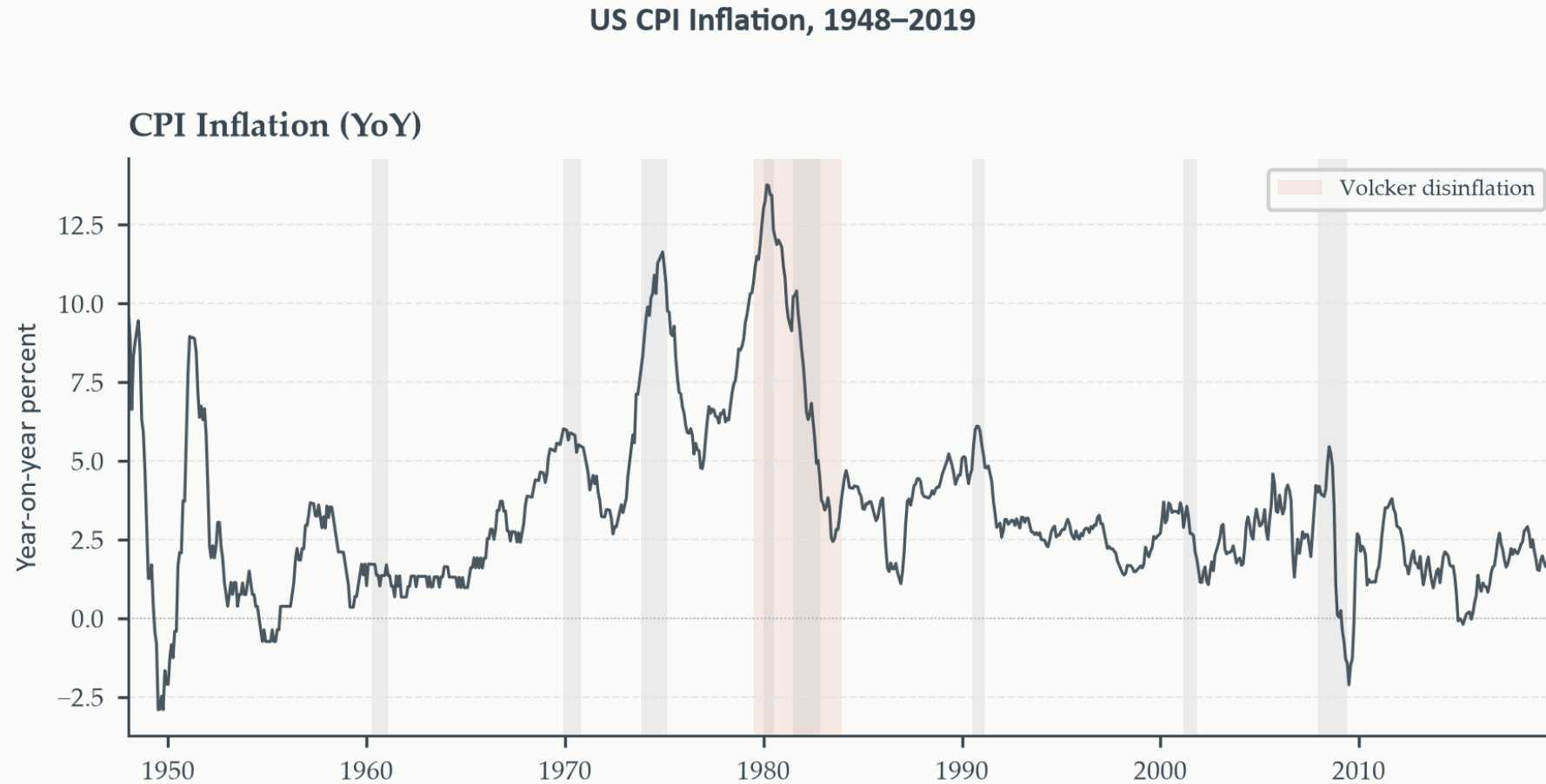
# Unit roots or broken trends?

Section 4

# Unit roots or broken trends?

- A broken trend can look similar to a unit root process
- US inflation
  - A standard ADF looking at 1947-2019 will see a slow upward drift from -3% to 13%, and then a downward drift back to 2-3%
  - The plot looks non-stationary
  - But, it can also be the case that the inflation series is combining two stationary series with different coefficients

# Unit roots or broken trends?



# Unit roots or broken trends?

- The Zivot-Andrews Model

- Model C

- $y_t = c + \beta t + (\gamma DU_t(\tau) + \delta DT_t(\tau)) + \alpha y_{t-1} + (\sum_{j=1}^k d_j \Delta y_{t-j}) + \varepsilon_t$

- $DU_t(\tau) = 1\{t > \tau\} \rightarrow$  level-shift dummy

- $DT_t(\tau) = (t - \tau)1\{t > \tau\} \rightarrow$  slope-shift dummy

- Test

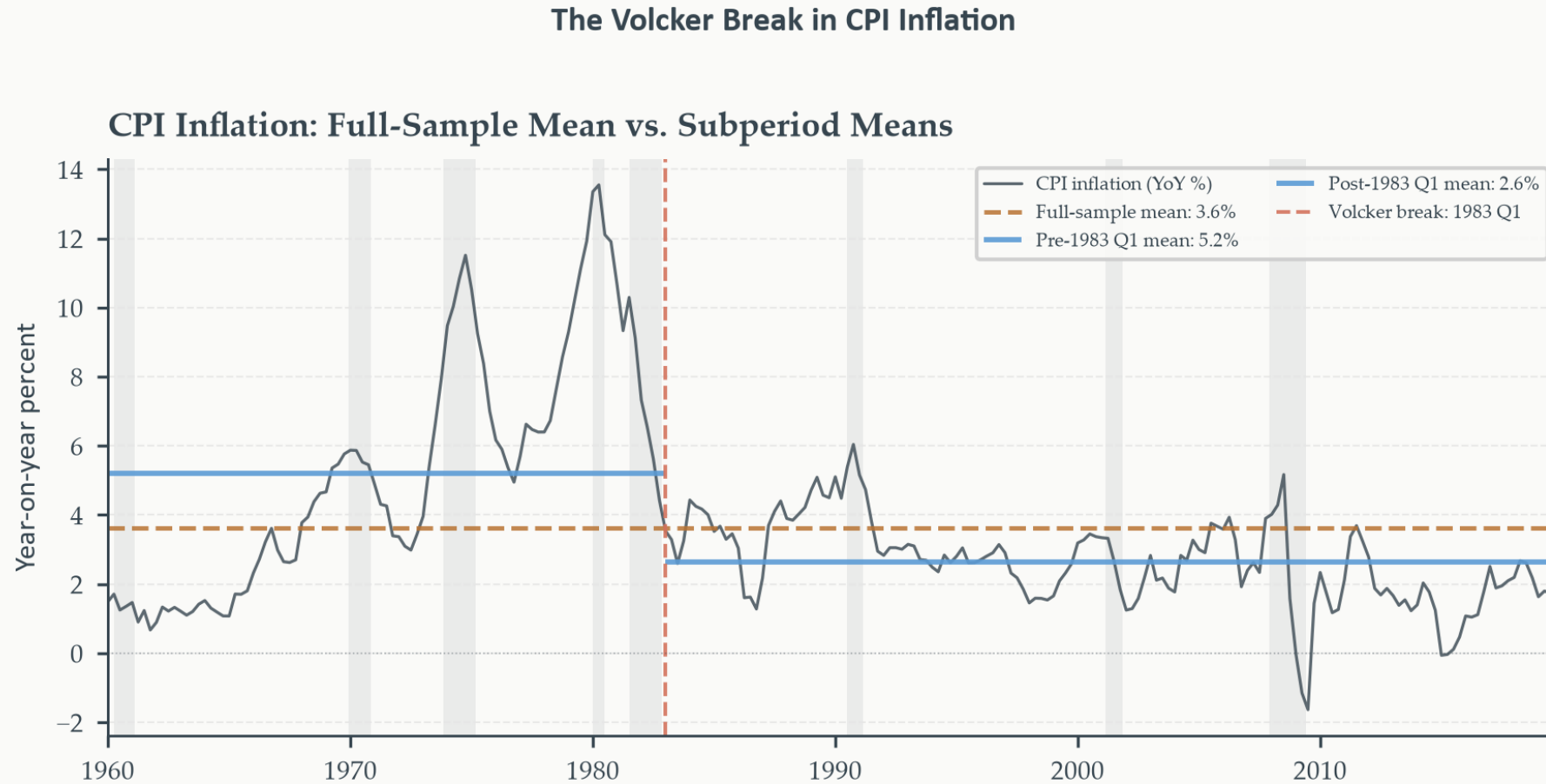
- $H_0: \alpha = 1$  (unit root)

- $H_a: |\alpha| < 1$  (stationary around a **broken** deterministic trend)

- $ZA = \inf_{\tau \in \hat{\tau}} t_{\hat{\alpha}}(\tau)$

- The break date  $\tau$  is endogenous; run the test for all candidates and keep the one that produces the **most negative**  $t$ -statistic on  $\hat{\alpha}$

# Unit roots or broken trends?



# Unit roots or broken trends?

- ZA limitation
  - The ZA test searches for all candidate breaks and selects the one that produces the most negative  $t$ -statistic
  - On **highly persistent series**, this tends to place the estimated break near the **trimming boundary** — not at the true economic break
    - Why? High AR persistence flattens the  $t$ -statistic profile across dates
    - The test will reject (or fail to reject) correctly
  - Practical implications
    - When economic theory identifies a known break, use it
    - This test is more reliable when the break is large, abrupt, and well inside the sample

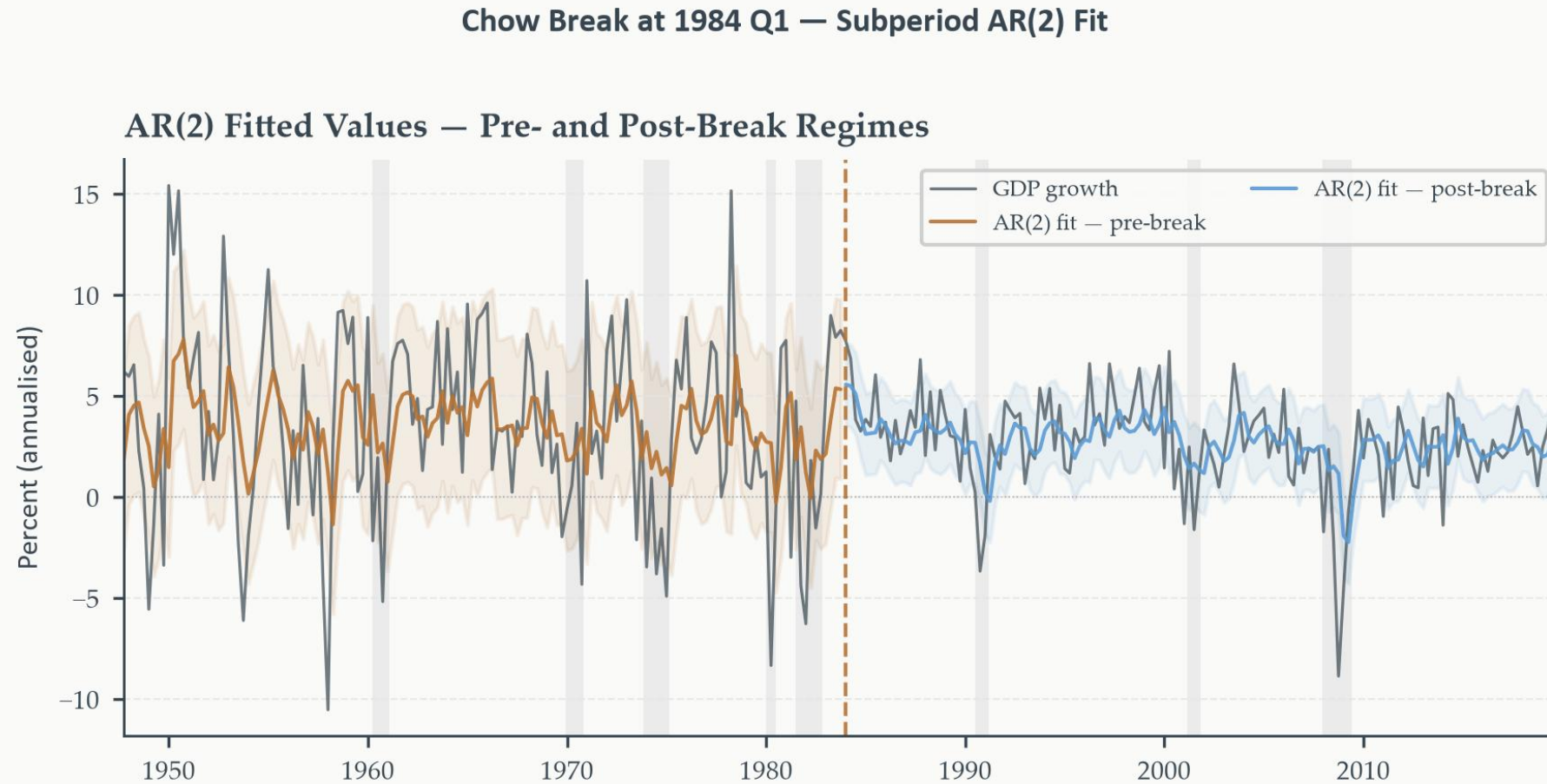
# How to react to structural breaks

Section 5

# How to react to structural breaks

- Option 1: Subperiod estimation
  - Split the sample on each  $\tau$  and estimate one independent model for each regime
  - Trade-off: A more relevant model per period, but less precision due to dropping data

# How to react to structural breaks



# How to react to structural breaks

Forecast Accuracy – Break Response Strategies

h = 1 quarter | Post-1984 Q1 evaluation | 40-quarter rolling window

| Strategy                       | RMSE  | vs Full-sample |
|--------------------------------|-------|----------------|
| Full-sample ARIMA(1,1,1)+drift | 2.285 | 1.000          |
| Post-break ARIMA(1,1,1)+drift  | 2.165 | 0.948          |
| ARIMA + level-shift dummy      | 2.282 | 0.999          |

RMSE in annualised percentage points.

Ratio: RMSE of strategy / RMSE of full-sample baseline.

Ratio < 1 → strategy beats the full-sample baseline.

# How to react to structural breaks

- Why not MAPE?

- $MAPE = \frac{1}{T} \sum_{t=1}^T \left| \frac{e_t}{y_t} \right|$

- Problem: GDP growth crosses zero during recessions
    - 2008-09 trough: annualized growth = -8.9%
    - $|y_t|$  near zero  $\rightarrow$  MAPE explodes
    - One recession quarter can dominate the entire sample average
- Instead, report RMSE and MAE

# How to react to structural breaks

- Option 2: Dummy variables

- Retain the full sample and add dummies to identify each period

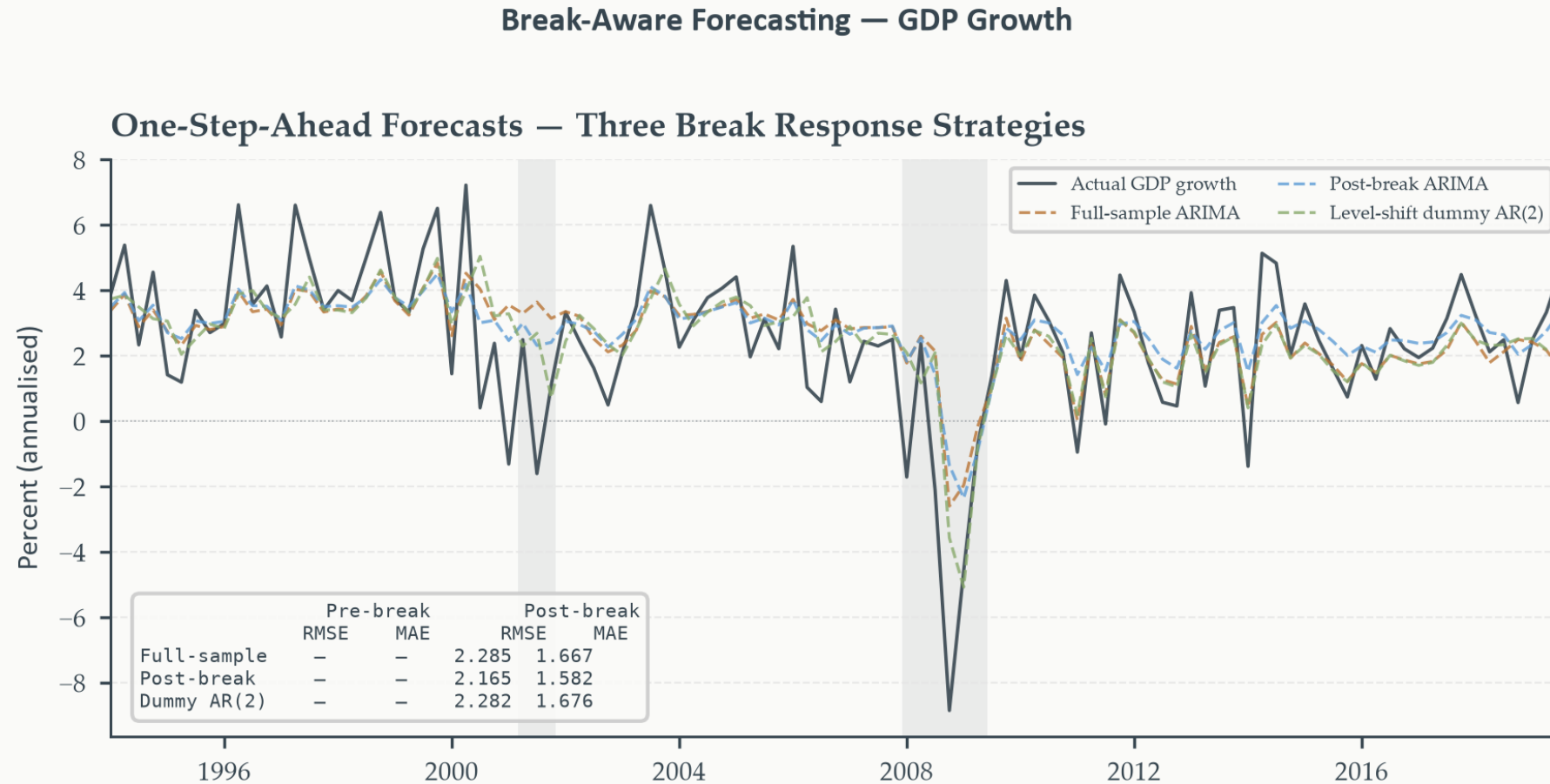
- $$y_t = \underbrace{(\bar{y} + \gamma D_t)}_{\text{mean break}} + \underbrace{(tD_t + \beta t)}_{\text{trend break}} + \underbrace{(\nu D_t y_{t-1})}_{\text{coeff break}} + \phi_1 y_{t-1} + \dots + \varepsilon_t$$

- Uses all observations → more efficient than subperiod estimation, but the model “shares” the same variance across structural breaks

# How to react to structural breaks

- Option 3: Split-sample forecasting
  - Use the full sample for estimation precision (as in Option 2)
  - Generate forecasts leaning on the post-break regime — or use a rolling window

# How to react to structural breaks



# How to react to structural breaks

- Option 4: Markov-Switching model
  - Particularly useful when the process switches back and forth
    - Expansions and recession
    - Calm and turbulent markets
  - Parameters follow an unobserved Markov chain: at each date, the process is one of  $K$  regimes, transitioning to a different regime according to a fixed probability matrix  $P$ 
    - $\hat{P} = \begin{pmatrix} 0.95 & 0.05 \\ 0.25 & 0.75 \end{pmatrix}$
    - 95% chance of staying in expansion and 5% chance of entering recession
    - 25% of recovering from a recession and 75% chance of staying in the recession
    - Expected duration of each regime:  $\frac{1}{1-p_{ii}}$

# Structural Breaks

Chapter 6

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